

Solution Requirements

Date	03 October 2026
Team ID	SWTID-2026-5999
Project Name	PocketSmart AI — AI-Assisted Goal-Based Budget Planner
Maximum Marks	4 Marks

Step 1: Functional Requirements

S.No	Requirement Category	Requirement Description	Priority
1	Account Management	Register and sign in with validated credentials; passwords are stored as secure hashes	High
2	Access Control	Protected planners and history are available only to authenticated users, and a user can access only their own saved plans	High
3	Budget Input	Accept a total INR budget together with planner-specific quantities, guest count, occasion or preferences	High
4	AI Recommendation	Send validated planning information to Gemini and receive a structured recommendation rather than unrestricted text	High
5	Budget Verification	Recalculate item totals on the server and remove items if necessary so the displayed plan remains within the submitted budget	High
6	Shopping Support	Generate category-appropriate search links so the user can continue the purchase research on external platforms	Medium
7	History	Save completed plans and allow the owner to open the detailed breakdown later	Medium
8	Reliability	Provide a deterministic fallback plan when the AI service is unavailable or returns unusable data	High
9	Image Support	Accept an optional JPG, PNG or WEBP outfit image for the jewelry planner, with size and type validation	Medium
10	Reporting	Show planned spend, remaining amount, category allocation and recommendations in a readable result view	Medium

Step 2: Non-Functional Requirements

S.No	NFR Category	Requirement	Acceptance Criteria
1	Performance	Application-side processing should remain lightweight before external AI latency	Repository load evidence shows sub-second average application overhead in offline mode
2	Security	Secrets must remain outside source code and protected routes must reject unauthenticated access	API key is read from environment; protected routes require a valid session
3	Reliability	AI failure must not leave a valid request without a usable response	Fallback plan is returned when Gemini is unavailable
4	Usability	A first-time user should understand the budget, inputs and result without technical knowledge	Responsive forms, clear labels, validation messages and visible totals
5	Maintainability	Planner-specific logic should be separated from authentication, storage and presentation	Dedicated modules for routes, models, AI service, authentication and database
6	Portability	The project should run on a normal development laptop	Python environment plus <code>requirements.txt</code> ; optional Docker support